

# Jonathan Wrede

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AI Platform / ML Infrastructure Engineer focused on Kubernetes ML systems, inference readiness, observability, and data platforms.

Built production workflows for 100,000+ IoT devices; OSS work in inference-readiness-kit, llmprobe, llm-d, OpenTelemetry, and Feast.

## EXPERIENCE

- Vilisto** Hamburg, Germany  
*Machine Learning Engineer* Jan. 2025 - Present
  - Battery health & runtime prediction:** Designed capacity and runtime prediction algorithms with confidence intervals for 100,000+ IoT thermostats (~400 customers); reduced battery replacement costs by EUR 50-100K in year one. Pipeline with **Dagster** and **MLflow**.
  - Company data platform (ELT):** Built the analytics warehouse: 95 **dbt** models ingesting 5 data sources with PII-safe pipelines, automated tests/lineage via **Dagster**, staged rollouts with ArgoCD.
  - Fleet monitoring:** **Grafana** dashboards and **Python** alerting service for 100,000+ devices, replacing days of manual checks with automated detection of offline thermostats and gateway failures.
  - Platform & tools:** Deployed and monitored data/ML services on **Kubernetes (Helm, probes, logs/metrics)**; shipped a **FastAPI** lab dashboard for battery test visualization with automated alerting.
- Saracus** Muenster, Germany  
*Data Platform Consultant* Jan. 2023 - Dec. 2024
  - Cloud data platforms:** Designed and deployed data warehouses for 5 enterprise clients in insurance, e-commerce, and energy on **Snowflake (AWS)**, **Google BigQuery**, and **Azure Synapse**; built automated **Python** ETL pipelines processing millions of records.
  - Application development:** Built multiple **Streamlit** apps with OAuth SSO, Snowflake integration, and custom visualization; deployed via **Azure Container Apps**.
  - Data governance:** Implemented a **Master Data Management** solution with **Ataccama + Azure**; automated glossary maintenance with **Azure Functions** and Python.
- Vilisto** Hamburg, Germany  
*Working Student - Data Scientist* Sep. 2021 - Oct. 2022
  - ML & data engineering:** Extended presence-detection models with **Python/scikit-learn** for thousands of IoT thermostats; built schemas and pipelines in **PostgreSQL/TimescaleDB** with **SQL** and **pandas**.
  - Software development:** Contributed to backend (**Python, Docker**) and frontend (**TypeScript, Vue.js**); built a **PyQt** labeling tool and led internal programming workshops.
- TU Dortmund** Dortmund, Germany  
*Teaching Assistant - Data Structures & Algorithms* Feb. 2021 - Oct. 2021

## EDUCATION

- University of Muenster** Muenster, Germany  
*Master Computer Science Focus AI / ML / Statistics* Oct. 2023 - Oct. 2026  
*Thesis: Memory profiling and deployability of privacy-preserving transformer inference systems*
- TU Dortmund** Dortmund, Germany  
*Bachelor Computer Science with Minor Statistics* Oct. 2019 - Apr. 2023  
*Thesis: Phase transitions in sparse principal component analysis*

## OPEN SOURCE & PROJECTS

- AI platform OSS PRs:** 7 merged infrastructure PRs across llm-d, OpenTelemetry, and Feast: router CI hardening, OpenAI telemetry, BigQuery/DynamoDB/Trino feature-store fixes.
- Transformer Inference Memory Benchmark:** M.Sc. thesis artifact for resource profiling and deployment feasibility of transformer inference under hard memory constraints; measured operator-level RSS, allocator behavior, and runtime tradeoffs across FHE/MPC/FSS systems; reduced peak memory by 61-99% in optimized variants. **Tech:** Python, C++, Docker, PyTorch
- inference-readiness-kit / llmprobe:** LLM inference readiness workflow: external probes, SLA gates with exit codes, Prometheus/vLLM correlation, concurrency sweeps, Grafana inspection, runbooks, tests, and CI. **Tech:** Python, Go, Prometheus, Grafana, vLLM

## TECHNICAL SKILLS

- Target AI Inference Stack:** vLLM, SGLang, KServe, Gateway API/Envoy, llm-d, OpenTelemetry, Prometheus/Grafana, readiness gates, KV cache, TTFT/TPOT benchmarking
- Infrastructure & Cloud:** Kubernetes (Helm, ArgoCD), Terraform, Docker, AWS, Azure, GCP, Linux
- Data & Backend:** Python, Go, SQL, dbt, PostgreSQL/TimescaleDB, Snowflake, BigQuery, FastAPI, Pandas/Polars
- Also used:** TypeScript, Streamlit, Vue.js, scikit-learn, Ataccama, MinIO, Java